Mississippi Valley State University

CS 341 — Discrete Structures

Lecture 14: Writing the Proof

Double Inclusion, Disproof, and Partitions

Chartrand & Zhang, Sections 2.2 and 2.4

Name: _____ Date: _____

Where we are

Wednesday you verified set identities with membership tables — complete, mechanical, and limited to a handful of sets. Today you learn the argument mathematicians actually write, and it is your first real proof. Everything in Chapter 3 builds on it.

By the end of today you should be able to

1. Prove $A = B$ by showing $A \subseteq B$ and $B \subseteq A$.
2. Disprove a proposed set identity with a counterexample.
3. Say what a partition of a set is.

1. Set equality is a biconditional

The double-inclusion principle

$$A = B \quad \text{if and only if} \quad A \subseteq B \ \text{ and } \ B \subseteq A.$$

This is Lecture 6 wearing set notation. To prove a biconditional you must prove two things — so a proof of a set identity has **two halves**, and stopping after one proves nothing.

Recall also that $A \subseteq B$ means $\forall x,\ (x \in A \Rightarrow x \in B)$. So each half is proved by:

The element-chasing template

Let $x \in A$. . . . reasoning . . . *Therefore* $x \in B$. *Hence* $A \subseteq B$.
The x is **arbitrary** — you assume nothing about it beyond membership. That is what makes the conclusion universal.

2. Worked together

Claim

For all sets A, B, C: $(A - B) \cup (A - C) = A - (B \cap C)$.

Half one: $(A - B) \cup (A - C) \subseteq A - (B \cap C)$.

Let $x \in (A - B) \cup (A - C)$. Then $x \in A - B$ or $x \in A - C$.

Case 1: $x \in A - B$, so $x \in A$ and $x \notin B$. Since $x \notin B$, certainly $x \notin B \cap C$.

Case 2: $x \in A - C$, so $x \in A$ and $x \notin C$. Since $x \notin C$, again $x \notin B \cap C$.

In both cases $x \in A$ and $x \notin B \cap C$, so $x \in A - (B \cap C)$.

Half two: $A - (B \cap C) \subseteq (A - B) \cup (A - C)$.

Let $y \in A - (B \cap C)$. Then $y \in A$ and $y \notin B \cap C$. Since y is not in both B and C, it fails to be in at least one of them.

If $y \notin B$ then $y \in A - B$. If $y \notin C$ then $y \in A - C$. Either way $y \in (A - B) \cup (A - C)$.

Combining the two halves, the sets are equal. ■

Where the logic came from

Half two used: *y is not in both B and C, so y fails at least one.* That is $\sim(P \wedge Q) \equiv (\sim P) \vee (\sim Q)$ — De Morgan, Theorem 1.24.
Every step of a set proof is a logical law you already have. The only new skill is **writing it in sentences**.

3. Disproving a proposed identity

Not every plausible-looking identity is true.

Claim to test

For all sets A and B: $(A \cup B) - B = A$.

Try to find a counterexample before reading on.

Your work

Take $A = \{1, 2\}$ and $B = \{2\}$. Then $A \cup B = \{1, 2\}$, so $(A \cup B) - B = \{1\}$. But $A = \{1, 2\}$. Since 2 belongs to A and not to the left side, the two sets differ. **The claim is false.**

What a counterexample must supply

Specific sets, and the **specific element** where the two sides disagree. "It doesn't always work" is not a disproof.

Note also what *is* true: $(A \cup B) - B = A - B$. The claim fails precisely on the part of A that overlaps B, and naming that is worth more than the counterexample alone.

The asymmetry, one last time

To **prove** an identity: an argument covering every element.
To **disprove** one: a single pair of sets and a single element.
Week 1 with $n^2 - n + 41$, Week 3 with satisfiability, Week 4 with quantifiers, and now here. It is the same fact each time.

4. Partitions

Definition — partition

A **partition** of a nonempty set A is a collection of nonempty subsets of A such that every element of A belongs to **exactly one** of them.

Equivalently, the subsets are pairwise disjoint and their union is A.

Which of these partition $A = \{1, 2, 3, 4, 5, 6\}$?

Collection	Partition?	Why
$\{1, 2\},\ \{3, 4\},\ \{5, 6\}$	Yes	Disjoint, and together they give A.
$\{1, 2, 3\},\ \{3, 4, 5, 6\}$	No	3 is in both — not disjoint.
$\{1, 2\},\ \{3, 4\}$	No	5 and 6 are left out.
$\{1, 2, 3, 4, 5, 6\}$	Yes	One block is allowed.
$\{1, 2\},\ \emptyset,\ \{3, 4, 5, 6\}$	No	Blocks must be nonempty.

Where partitions go next

The even and odd integers partition $\mathbb{Z}$. Sorting objects into groups where every object lands in exactly one group is the shape of an **equivalence relation**, which is the next unit.

Exit ticket

1. What two things must you show to prove $A = B$?

2. Give a counterexample to: *for all sets A and B, $A - B = B - A$.*

Before Monday

Read Section 2.4, "Partitions of Sets," and review the Chapter 2 Highlights on p. 75–76. The Key Results page collects every law from this week on a single sheet.

Self-check (not collected). Section 2.2, Exercises 17, 19, and 21 — odd-numbered, answers in the back.

Coming up

Homework 5 is due tonight, Friday, September 18, 11:59 p.m.
Quiz 5 is Monday, September 21 — 15 minutes on Section 2.2.
Homework 6 covers Sections 2.2 and 2.4, due Friday, September 25.

Questions I still have